Supplementary Material for Knowing When Not to Transfer

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This supplementary file records the reproducibility boundaries of the TKDE revision. The executable source, configurations, raw outputs, aggregation scripts, and the complete appendix are distributed with the main manuscript.

I. Protocol and Information Boundary

All primary ranking experiments use chronological 70/15/15 splits and collision-filtered historical negatives. HRCSR reserves a chronological suffix of the target-training period for calibration. The checkpoint-boundary audit exposes validation-selected, fixed-epoch, and optimization-prefix-only modes; a mode is reported only when its raw JSON output is present.

The completed matched-width audit is summarized in the main paper. The earlier reduced-width diagnostic is retained below only to make the version boundary auditable; it is not used for a primary claim.

TABLE I
CHECKPOINT-BOUNDARY DIAGNOSTIC (REDUCED-WIDTH AUDIT CONFIGURATION).

Target	Selection	Hard MRR	Route−Id.	Open
Wikipedia	validation	.4551(.0572)	+.0012	1/5
Wikipedia	fixed epoch	.4510(.0498)	+.0016	1/5
Wikipedia	prefix only	.4510(.0498)	+.0016	1/5
Reddit	validation	.4296(.0121)	+.0002	1/5
Reddit	fixed epoch	.4306(.0137)	+.0002	2/5
Reddit	prefix only	.4306(.0137)	+.0002	2/5
MOOC	validation	.4559(.0182)	+.0742	5/5
MOOC	fixed epoch	.4567(.0205)	+.0802	5/5
MOOC	prefix only	.4572(.0194)	+.0799	5/5

This diagnostic is not substituted for the main table: it uses the released reduced-width audit configuration and therefore only tests sensitivity to checkpoint-selection information within that branch. The matched-width rerun, including UCI and Enron, is recorded in `CHECKPOINT_BOUNDARY_RESULTS.csv` and the experiment manifest.

II. Sparse SocialEvo Boundary

SocialEvo now has complete five-seed HRCSR, strict-RASP, all-source, target-only T-GAT, and target-only GraphMixer artifacts for seeds 0, 1, 2, 3, and 42. Direct per-seed transfer JSON files are authoritative because an older resumed-run aggregate retained only seeds 3 and 42.

The SocialEvo runs use the audited skip-unsampleable sparse-negative policy. Across the common processed split, validation retains 307,953 of 314,930 requested queries and test retains 310,119 of 314,924; no random fallback is used. Route, identity, hard, and soft paths share the same queries and historical negatives. The target-only summaries record the chronological split, historical ranking, and 20 negatives but not per-query identifiers, so they are contextual controls rather than proof of exact cross-process query-ID equality. The main paper consequently reports SocialEvo in a separate five-seed sparse-domain table instead of pooling it with the dense panel.

III. Contemporary Baseline Compatibility

MiNT, BRIDGE, and GraphBridge are discussed in the main paper and audited in `CONTEMPORARY_BASELINE_AUDIT.md`. Their released tasks or metrics do not permit a direct historical-negative MRR comparison without changing the scientific question; no cross-task numbers are imported.

IV. Measured Efficiency

The common-hardware measurements and synthetic expert-count scaling profile are provided in `EFFICIENCY_RESULTS.csv` and `EXPERT_SCALING.csv`. The latter is a computational profile using synthetic expert replication and is not a performance benchmark.

V. Result Provenance

Every manuscript number is linked in `FINAL_RESULT_PROVENANCE.md` to a raw JSON/CSV file and an aggregation script. Unavailable fields remain explicitly marked `NA`; no value is imputed.

VI. Reproduction

The complete command sequence is in `RUN_ALL.md`. Use `scripts/audit_sparse_protocol.py` for query-denominator checks and `scripts/profile_efficiency.py` for measured cost fields.